The FOR-NEXT loop between lines 340 and 580 controls how the 16000 segments are processed. In line 360, a subroutine loads the next segment to be processed into XX[0] through XX[624], and sets XX[625] through XX[1023] to a value of zero. In line 370, the FFT subroutine is used to find this segment's frequency spectrum, with the real part being placed in the 513 points of REX[ ], and the imaginary part being placed in the 513 points of IMX[ ]. Lines 390 to 430 show the multiplication of the segment's frequency spectrum, held in REX[ ] & IMX[ ], by the filter's frequency response, held in REFR[ ] and IMFR[ ]. The result of the multiplication is stored in REX[ ] & IMX[ ], overwriting the data previously there. Since this is now the frequency spectrum of the output segment, the IFFT can be used to find the output segment. This is done by the mythical IFFT subroutine in line 450, which transforms the 513 points held in REX[ ] & IMX[ ] into the 1024 points held in XX[ ], the output segment.

Lines 470 to 550 handle the overlapping of the segments. Each output segment is divided into two sections. The first 625 points (0 to 624) need to be combined with the overlap from the *previous* output segment, and then written to the output signal. The last 399 points (625 to 1023) need to be saved so that they can overlap with the *next* output segment.

To understand this, look back at Fig 18-1. Samples 100 to 199 in (g) need to be combined with the overlap from the *previous* output segment, (f), and can then be moved to the output signal (i). In comparison, samples 200 to 299 in (g) need to be saved so that they can be combined with the *next* output segment, (h).

Now back to the program. The array OLAP[ ] is used to hold the 399 samples that overlap from one segment to the next. In lines 470 to 490 the 399 values in this array (from the previous output segment) are added to the output segment currently being worked on, held in XX[ ]. The mythical subroutine in line 550 then outputs the 625 samples in XX[0] to XX[624] to the file holding the output signal. The 399 samples of the current output segment that need to be held over to the next output segment are then stored in OLAP[ ] in lines 510 to 530.

After all 0 to 15999 segments have been processed, the array, OLAP[ ], will contain the 399 samples from segment 15999 that should overlap segment 16000. Since segment 16000 doesn't exist (or can be viewed as containing all zeros), the 399 samples are written to the output signal in line 600. This makes the length of the output signal $16000 \times 625 + 399 = 10{,}000{,}399$ points. This matches the length of input signal, plus the length of the filter kernel, minus 1.

Speed Improvements

When is FFT convolution faster than standard convolution? The answer depends on the length of the filter kernel, as shown in Fig. 18-3. The time